6.100B Recitation 1 🍔🍺🍕

Please join the Slido and answer the poll:

slido.com - code 2625862

Announcements!

- First microquiz on Tuesday, Feb 17
  - First ~35 minutes of the timeslot
  - No lecture after that
  - **You may bring one double-sided page of notes with you**
  - No other resources; *no IDEs* beyond the built-in code editor

- Problem Set 1 due next Wednesday at 9pm
- Come to office hours if you need help!!
  - M-W 5-9 pm, F 3-7 pm (all in 38-370)

What is the point of today?

- Review material from the last few lectures

- Prep you for the micro-quizzes
  - Maybe coding if we have time?

- Learn things have fun :)

Today we might cover

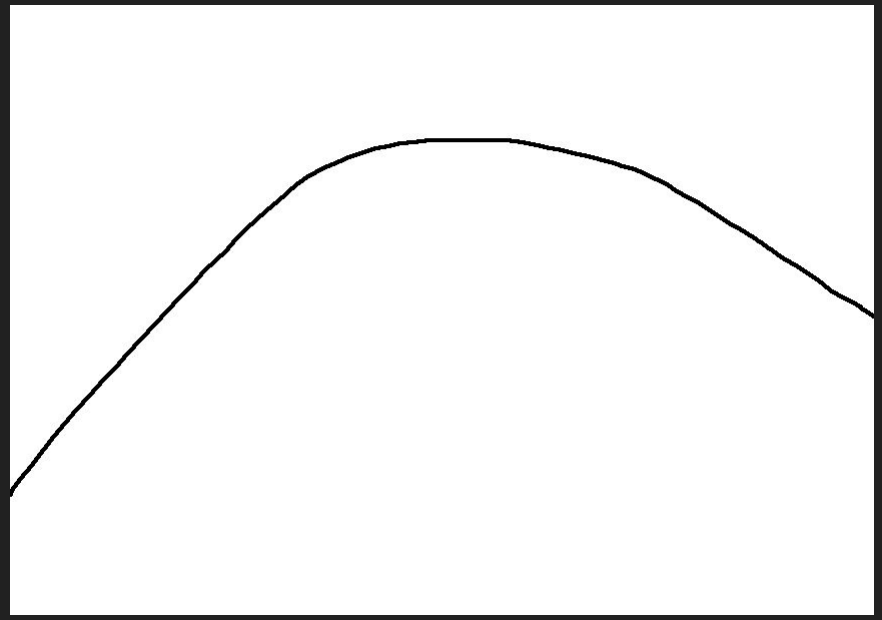
- What are optimization problems again?
- Ways to solve them
  - Ternary search
  - Greedy
  - Brute-force / decision tree
  - Dynamic programming

Today we might cover

- Some examples
  - Spring break trip
  - Revisit knapsack problem
  - House robbing problem

Spring break trip

Ternary search



What if the objective function was instead *concave* (i.e. going up, then going down)?

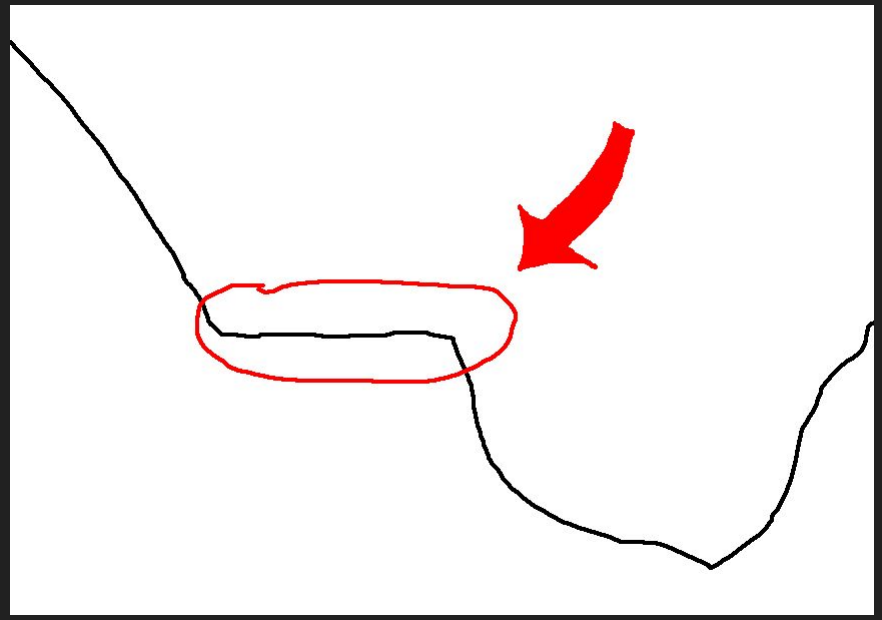
Your turn!

```python
def ternary_search(objective_fcn, min_val, max_val, precision):
    low, high = min_val, max_val
    # while high and low are too far apart
    while high - low > precision:
        guess1 = low + (high - low) / 3
        guess2 = high - (high - low) / 3
        print(f'min = {guess1:.4f}, max = {guess2:.4f}')
        if objective_fcn(guess1) > objective_fcn(guess2):
            low = guess1 # discard lower third
        else:
            high = guess2 # discard upper third
    return (low + high) / 2
```

Ternary search

What if the objective function had level parts?

Ternary search



What if the objective function had level parts?

Your turn!

Will the code provided in lecture work?

- Always
- Sometimes
- Never

Small Knapsack Example

capacity = 4 kg

$10
1 kg

$20
2 kg

$25
3 kg

What is our constraint? What are we maximizing?

DP =

When does memoization help?

- You can figure out the answer to the problem by using the answers to smaller versions of the problems

- We want the answers to these smaller versions of the problems more than once

When does memoization help?

- You can figure out the answer to the problem by using the answers to smaller versions of the problems
  “Optimal substructure”
  ??????????

- We want the answers to these smaller versions of the problems more than once
  “Overlapping subproblems”

How to memoize

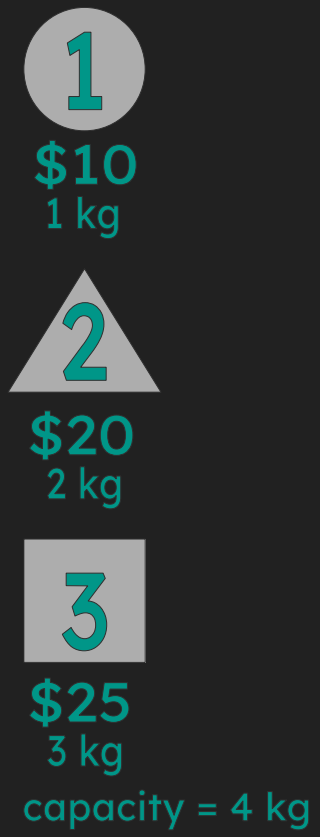
- **Use a table (tabular)**
  - Iterative
  - Considering all subproblems
  - “Bottom-up” Approach
- **Use a dictionary (memo)**
  - Recursive
  - Considering necessary subproblems
  - “Top-down” Approach

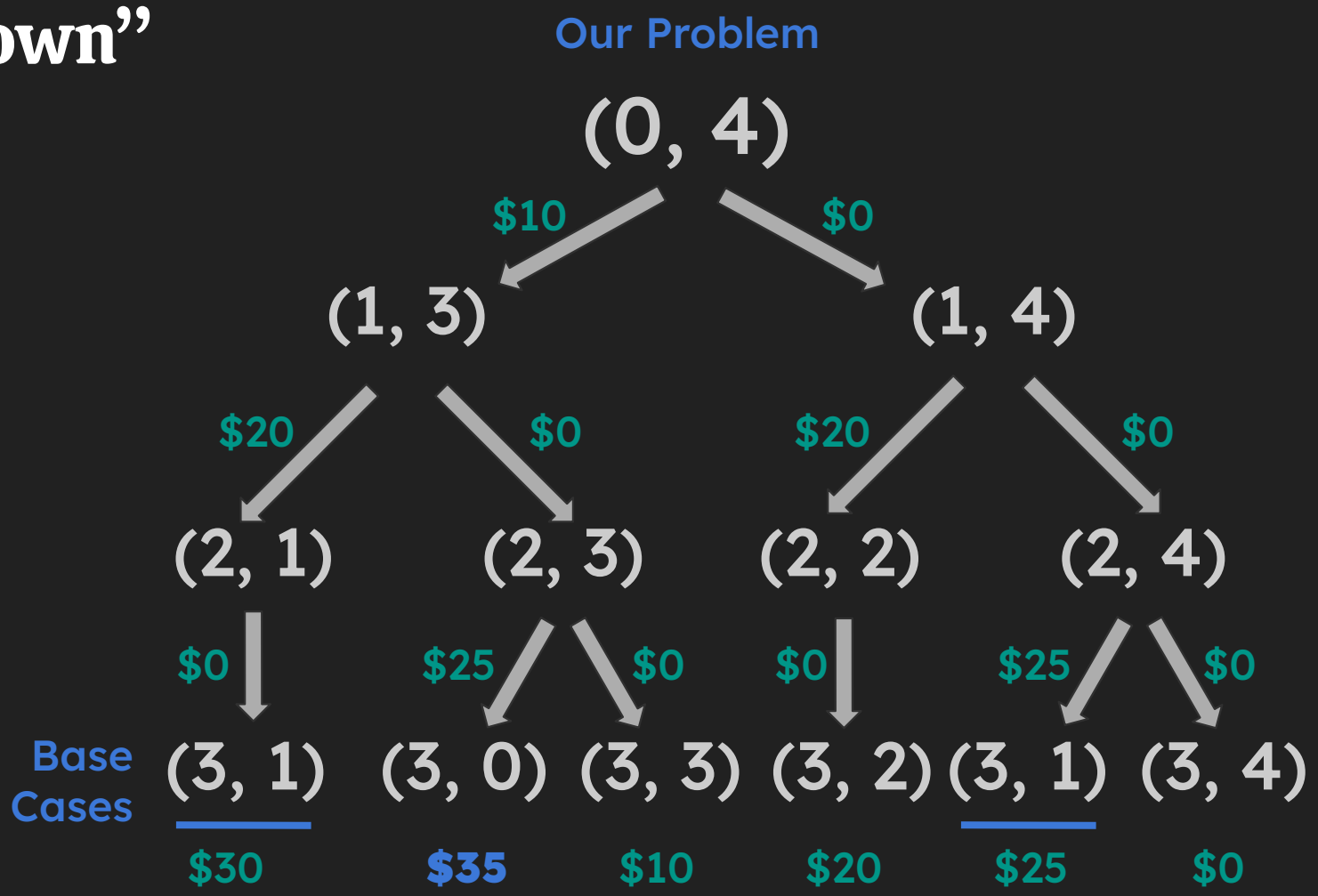
How do we define our subproblems?

(Index of first item left to consider , Remaining capacity in the knapsack)

capacity = 4 kg

"Top-Down"


1
$10
1 kg
2
$20
2 kg
3
$25
3 kg
capacity = 4 kg


Our Problem
(0, 4)
$10
$0
(1, 3)
(1, 4)
$20
$0
$20
$0
(2, 1)
(2, 3)
(2, 2)
(2, 4)
$0
$25
$0
$0
$25
$0
Base Cases
(3, 1)
(3, 0)
(3, 3)
(3, 2)
(3, 1)
(3, 4)
$30
$35
$10
$20
$25
$0

“Bottom-Up”

1	2	3
$10	$20	$25
1 kg	2 kg	3 kg

capacity = 4 kg

Bag Capacities

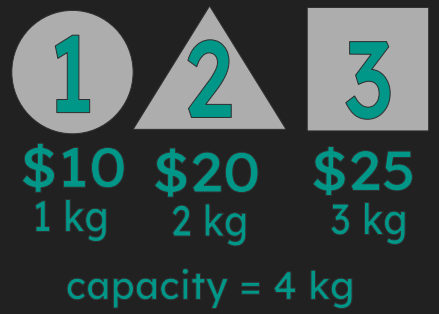
Index of item to consider

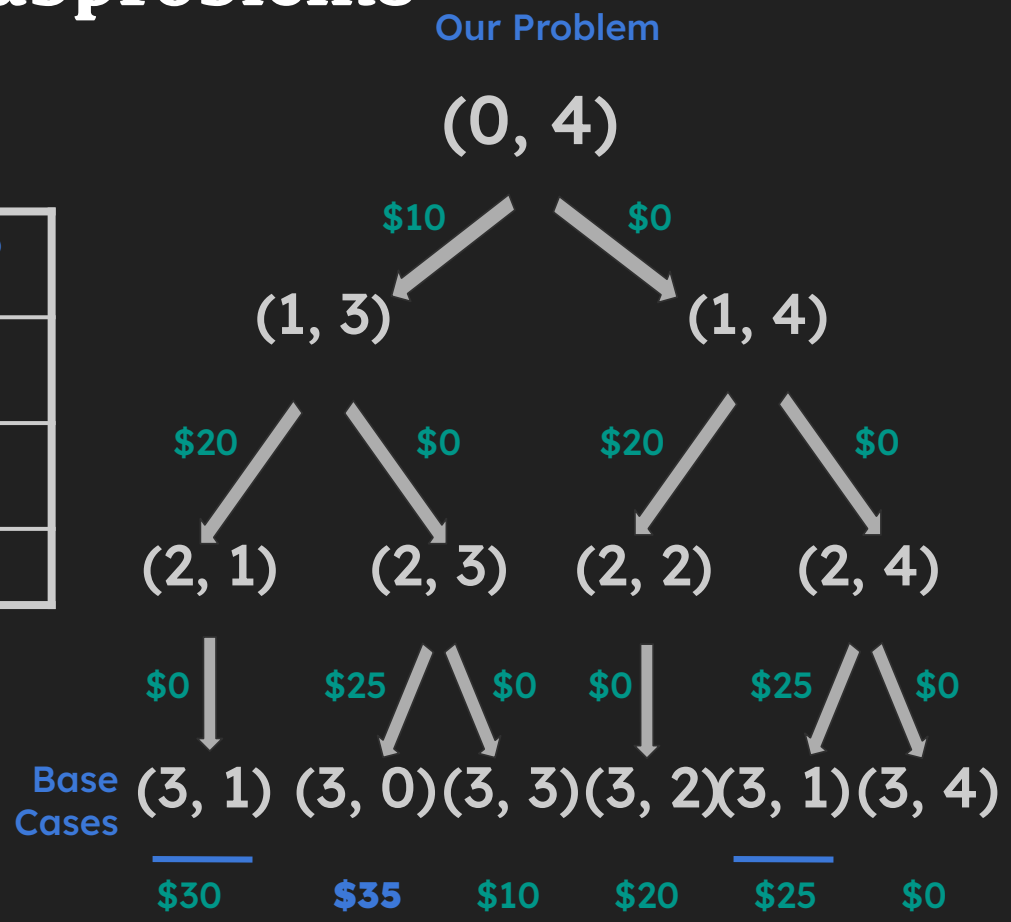
	0	1	2	3	4
0	0	10	20	30	**35**
1	0	0	20	25	25
2	0	0	0	25	25
3	0	0	0	0	0

They solve the same subproblems*

Bag Capacities

Index of item to consider	0	1	2	3	4
0	0	10	20	30	**35**
1	0	0	20	25	25
2	0	0	0	25	25
3	0	0	0	0	0

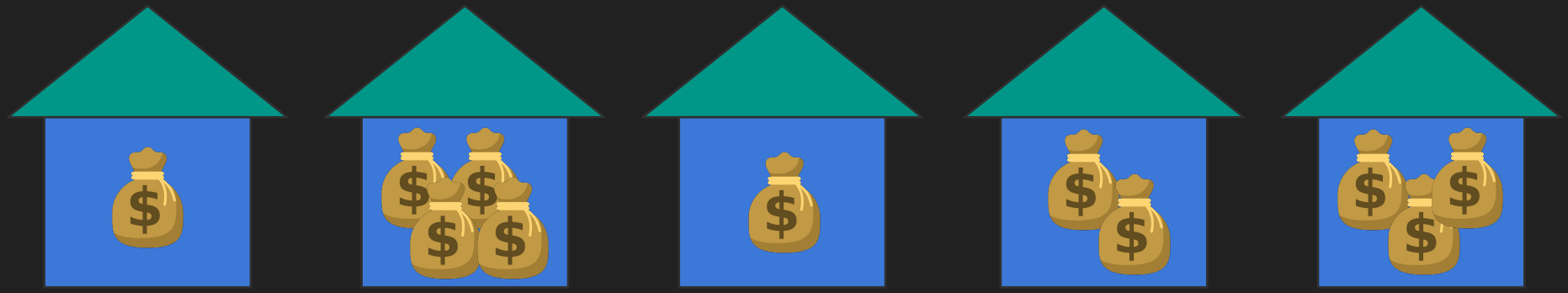




Complementary knapsack, “view the problem from the other side”

- “Min number of votes to gain so I win the election”
- “Max number of votes to keep so you get below the election threshold”

Problem: robbing houses



Problem: robbing houses

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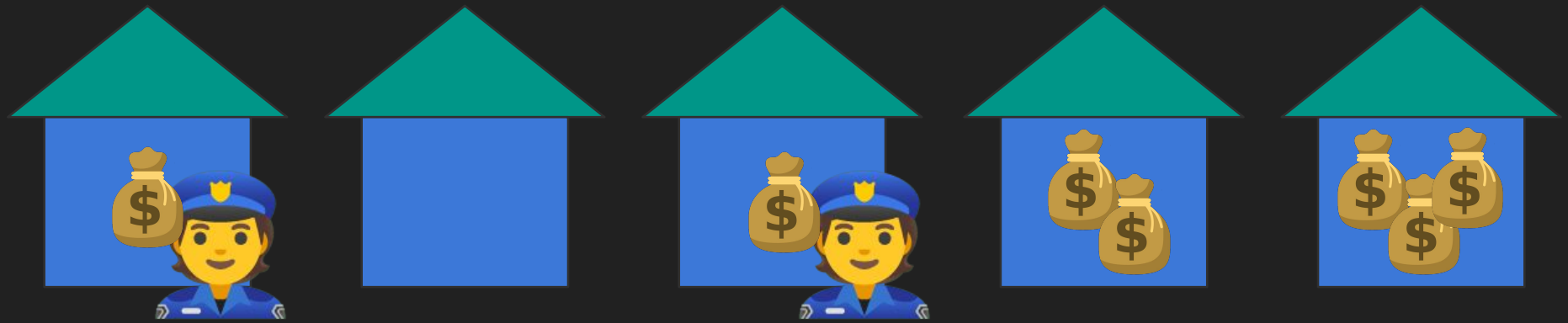


What's the maximum amount you can steal?

DISCLAIMER

THE SITUATIONS DISCUSSED ARE FOR ENTERTAINMENT PURPOSES ONLY. WHILE THIS LECTURE MAY PORTRAY CHARACTERS ENGAGING IN ACTS OF THEFT OR OTHER ILLEGAL ACTIVITIES, THE 6.100B STAFF DO NOT CONDONE, ENDORSE, OR ENCOURAGE SUCH BEHAVIOR IN ANY FORM. THE ACTIONS DEPICTED ARE INTENDED SOLELY FOR DRAMATIC STORYTELLING AND SHOULD NOT BE INTERPRETED AS PROMOTING OR GLAMORIZING CRIMINAL CONDUCT. STUDENTS ARE REMINDED THAT THEFT IS A SERIOUS OFFENSE WITH REAL-WORLD CONSEQUENCES, AND WE STRONGLY SUPPORT ETHICAL CONDUCT IN ALL CIRCUMSTANCES.

Will greedy work?

Let’s go rob one house: going back to knapsack

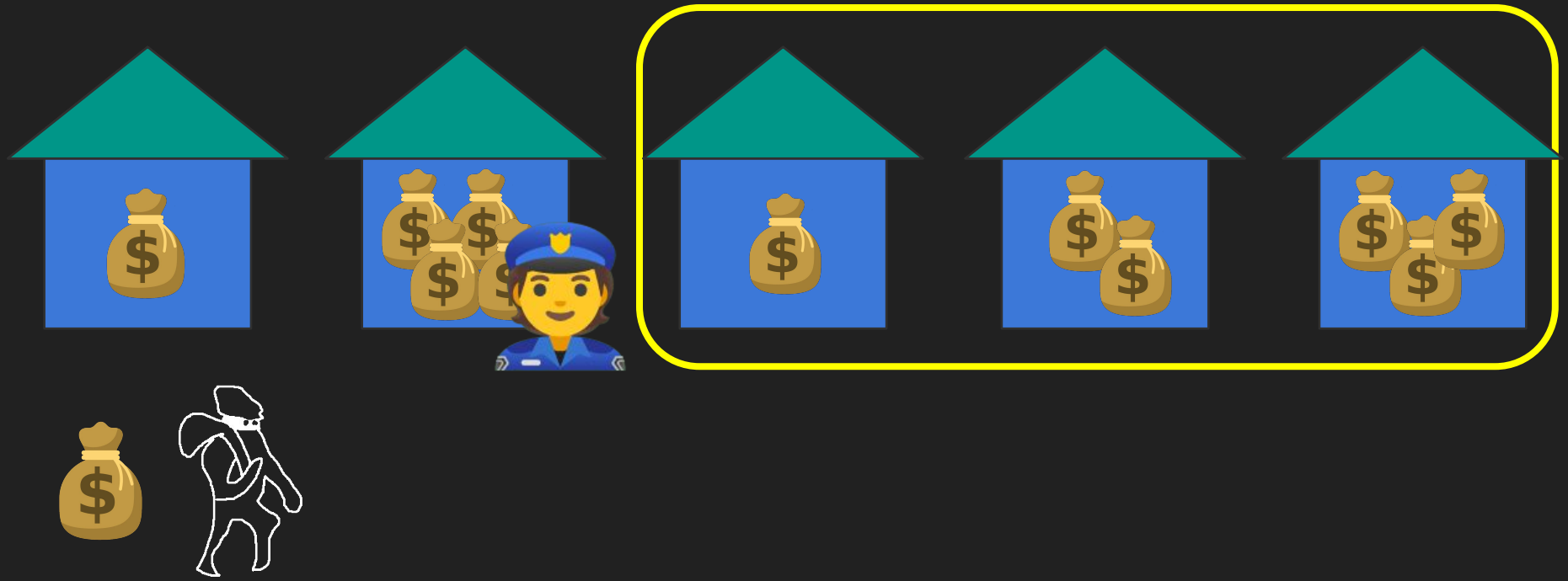
- What are the “smaller versions” of the problem?
- What’s their difference? What do they hinge on?

- “Is thing 1 included or not”

Let’s make a plan to rob all the houses

skip!

Let’s make a plan to rob all the houses



Want extra practice? Your turn!

Please download the coding exercises in the 6.100B site, and fill in the function. You may want to think about:

- The base cases
- The answer when the first house is broken into
- The answer when the first house isn’t broken into

Questions?

Thank you for attending :)

Come to office hours if you need help!!!

(M-W 5-9 pm, F 3-7 pm in 38-370)

6.100B

Assignments ▾ About ▾ Help ▾ Log In

- Help Queue
- Piazza 6.100B
- Extra Coding Practice
- MITx 6.0002 (optional finger exercises)
- Programming Resources
- General MIT Help

Intro to Computational Thinking & Data Science (Spring 2026)

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Announcements

ALL PREVIOUS ANNOUNCEMENTS

Lectures

Mon and Wed
11-12:30pm in 26-100
Materials in calendar below

Office Hours in 38-370

Start Feb 4
Mon,Tue,Wed: 5-9pm
Fri: 3-7pm

AI Tutor for independent learning

This Week's Pset

Problem Set 1
Due: Feb 18 at 09:00PM

Monday	Tuesday	Wednesday	Thursday	Friday
2/2: Lecture 1	2/3	2/4: Lecture 2	2/5	2/6